

Landslide Sentinel v0.1

Technical account of the accepted synthetic demonstration

Radiolaria OS / Hedgehog OS Source commit: 2f328be634247be11bc18a3b22a919f393d6ed1d **Date:** 15 September 2026 **Repository:** AAkhtanin/hedgehog-os

This appendix explains what the implementation does, how the retained observations support its claims, and which conclusions are outside the demonstrated scope. It is a new explanatory derivative. Source admission, historical execution, live model contribution, physical-domain fixtures and safe offline replay remain distinct forms of evidence.

The headline result is specific: changing the bounded semantic question changed the actual diagnostic work while common current-source and Root boundaries remained in force. A wider controlled story demonstrates local monitoring continuity across changed rainfall, optional reasoning delay, report delivery and measured recovery. No physical landslide forecast is claimed.

1. Source hierarchy and the meaning of acceptance

Landslide Sentinel v0.1 demonstrates one local cell retaining responsibility as observations, connectivity, reasoning roles and monitoring requirements change. The demonstrated physical domain is synthetic. The implementation uses real common C/D/E/Work/Root boundaries, real Gemini transport in a separately identified lane and historical real Environment Agency acquisition. Siren, report and local acquisition effects are admitted local mocks. The source and evidence landing is committed at 2f328be634247be11bc18a3b22a919f393d6ed1d. The radiolarian cover is conceptual domain artwork, not a sensor image or a scientific model. This presentation explains the scoped accepted demonstration, including preserved failures and evidence limitations.

The complete chronological story is CONTROLLED_SYNTHETIC: deterministic semantic fixtures and mock physical effects still pass real core machinery. The latest live lane contains five genuine Gemini requests, three roles in one C/D/Work/Root composition, a note-only semantic contrast and an internal-role emulation after site-network withdrawal. EA acquisition is a third lane: actual historical HTTP readings, including zeros, with interval and physical mapping limitations retained. These lanes support different claims and are not retroactively merged into a single all-live end-to-end run. The source-impact index bridges historical evidence to changed final bodies within named scope. Later landing tests and replay have separate receipts. A green label must never erase that distinction.

The current source tree is the landing commit above, whose parent is 50ab3916bff55e8034cf7e6c509d4803c5447589 and whose tree is 8db682205049c72c85500ebafade0696532338ef. A source-bound historical result is not relabelled as a fresh execution on that final tree. The accepted source-impact index identifies which changes affect a claim and which exact retained bodies or tests support continuity.

Evidence level	What it establishes	What it does not establish
Committed source + admission	Exact landed bodies and accepted repository relationship	A new physical experiment
Live capture + actual consumer	Model contribution consumed by recorded Work and Root	Universal model reliability
Controlled story	Core behavior with declared synthetic inputs and mocks	Real geotechnical truth

Evidence level	What it establishes	What it does not establish
Historical EA acquisition	Actual returned records and normalization	Applicability to the synthetic slope
External pin + safe replay	Exact supported derivative and relationships	Sensor certification or new action permission

The detached return is [RADIOLARIA_LANDSLIDE_SENTINEL_MAIN_LANDING_RETURN_20260915T070849Z.tar.gz](#), SHA256 [11e6b98a58420bc2e7fcf7a30b1bcbab40eea3fd52812db7dfd80fbe7042a1a6](#). It contains 472 files, including 471 manifest rows. This archive identity belongs to the return, not to the runtime proof package.

2. Local responsibility and the whole monitoring task

The task is not to have a chatbot announce a landslide. It is to keep a bounded local monitoring process coherent across changed evidence and optional dependencies. A synthetic slope supplies rainfall, displacement and an independent reserve channel. The site coordinates a finite catalogue of diagnostic operations and monitoring actions. It can ask an analyst which investigation would resolve the present uncertainty, but that response does not set thresholds, invoke an arbitrary sensor, modify the Host or authorize a signal. The chronological S0-S5 story exercises configuration, changed rainfall, roles, outage and signal, queued report delivery, measured recovery and local semantic memory. Separate latest live experiments establish real semantic producer-to-consumer relationships that the controlled whole story does not claim to demonstrate.

The architecture separates a semantic role from the physical engine that supplies its contribution. A local algorithm, a future local SLM or a cloud model can fill a permitted role with truthful provenance. Sentinel uses common semantic-work DTOs and trust profiles to validate a contribution, common BSEP/router boundaries for current meaning, actual Work execution and independent Root review. Root is neither the Gemini response nor the deterministic diagnostic function. A Root acceptance of information does not automatically create permission. When an effect is needed, a separate current action packet, live Host state and admitted capability validation are required. The point is local responsibility with replaceable reasoning, not a claim that the demonstrated software is a certified landslide-warning system.

The local Root identity persists across the relevant episode. A fresh invocation may be created for new work without manufacturing a second effect owner or resetting an existing task's spending. The domain adapter supplies a finite interpretation of observations and commands. It does not replace the shared semantic-work validator, execution-mode router, D/E runtime, Work materializer, Root decision kernel or effect firewall.

This distinction makes an edge deployment coherent: the cell can continue a bounded local responsibility even if a useful external analyst is unavailable. The current demonstration uses synthetic local observations and mock effect adapters. It shows that the program structure can maintain those boundaries; deployment still needs the physical and operational contracts appropriate to actual equipment.

3. Chronological S0-S5 behavior

The retained LS1 and LS2 histories provide the chronological baseline. S0 applies an initial configuration, performs a new calm intake and consumes local semantic memory. S1 changes rainfall while a next configuration is prepared but not yet executed. The old pending action is invalidated; actual public D/E recomputation returns changed work, retained sibling evidence and consumed results. A new current review and dispatch apply the next cadence. S2 consumes three controlled role proposals with independent current Work and Root checks. Controlled role fixtures are deliberately separated from the latest actual Gemini demonstration. The historical story includes real kernel execution; its source-bound measured times are not a physical sampling SLA.

S3 starts an optional reasoning worker before new critical local evidence arrives. The local task completes ordinary Work, Root review and native mock ON before the pending response returns; duplicate execution is

refused. S4 queues an immutable report through the local mock outbox and later records an exact receiver acknowledgement. S5 requires a measured low-value interval on both independent local channels and a distinct native OFF. The story closes with informational LocalDRS consumption. Late model text, a process cleanup, a return to network availability or a report acknowledgement cannot substitute for recovery evidence. These are related but distinct state transitions, each with its own identities and current boundary checks.

State	Input or change	Actual consequential boundary
S0	Calm current observations	Apply initial cadence and perform next intake
S1	Rainfall changes pending preparation	Refuse stale packet; consume E; apply new configuration
S2	Three bounded role proposals	Review work and admit any required reserve read
S3	Optional request remains pending; local conditions change	Current Root/Host complete local mock ON
S4	Report queued while delivery is unavailable	Exact message/hash acknowledgement after retry
S5	Fresh low-value recovery window	Separate current native OFF and informational memory

The story and the latest live experiment are complementary. The retained whole story has no paid model calls; the live lane demonstrates actual Gemini semantics in the current bounded consumer. The presentation does not replace one classification with the other. The story's incident becomes ACTIVE at scenario tick 150 and CLOSED at tick 255 in the cited LS2 record. These ticks are controlled domain time, not a measured real-world hazard response interval.

4. Typed observations and source meaning

One SiteSafetyRoot and persistent Host own a session. The domain adapter provides finite admitted transformations and effect operations. The semantic catalogue contains reference integrity, independent reserve series, temporal context applicability, spatial context applicability and evidence gaps. The native command interface includes monitoring configuration, an admitted reserve read, signal actions and report handling. Capability availability is specific to a source or transport: site-cloud context and remote rainfall may disappear while the separately authorized harness transport remains available. No generic execute-code, arbitrary source URL, policy-writing model capability or public emergency endpoint is part of this demonstration. The topology here is a conceptual explanation of the source-bound cell, not an exact rendering of every native graph artifact.

Typed observations bind raw references and hashes, site, channel, unit, statistic, measured interval, receipt time, sequence, calibration and lineage. C02 distinguishes an old event arriving now from a current observation and handles a same-event correction through actual applicability Work and Root. C03 records LIMITED with a missing primary and INSUFFICIENT when the reserve is also absent, without fabricated ON. C04 covers chronology/projection, duplicate/reorder behavior and source independence. Additional C08 evidence admits synthetic channel-level and vibration inputs under the same site Root. These finite checks establish the demonstrated measurement contract, not real sensor calibration, independent physical hardware or an externally certified observability metric.

The current observation representation carries `observation_id`, `raw_ref`, `raw_sha256`, `site`, `channel`, `value`, `unit`, `statistic`, `observed_start`, `observed_end`, `received`, `sequence`, `calibration`, `lineage`, `status` and a projection binding. Compatibility measurement frames remain narrower fixed-point records. Their equality is checked where used; they do not replace the richer evidence needed for chronology and independent witnesses.

Displacement and reserve use integer micrometres. Rainfall uses integer `mm_x1000`; the value 41000 corresponds to 41 millimetres in the synthetic policy example. Local critical and recovery checks require adequate current records, independent lineage identifiers and coincident measurement support. Independent identifiers within a fixture are demonstrated contractual independence, not a laboratory proof that two physical devices fail independently.

A new receipt timestamp cannot change when the measured event occurred. A correction to an event remains tied to its own raw and projection identity. Missing or inadequate data is a legitimate state that changes the permitted interpretation and action; it is not replaced by a normal value.

5. Duty-specific context and the model interface

`semantic_adapter_v01.context` filters channels by intended duty. Local slope and adversarial duties receive displacement and reserve observations plus the maintenance note. The environmental duty receives rainfall and omits that local note. Source records include identity, time, unit, calibration and lineage as applicable. The supplied context states the synthetic origin and the channels not included. It supplies duty-specific investigation contracts: target, intended use, required evidence, bounded result classes and limits. The model is asked to select a useful investigation, not to choose an outcome. Provider requests are captured privately with exact context digests. This is a finite semantic boundary, not a claim that the LLM never sees domain measurements or that all context is cryptographically secret.

The exact allowed response has diagnostic, `source_refs` and reason. `validate_response` checks a closed dictionary shape, an allowed diagnostic for this duty, a nonempty unique set of references belonging to supplied sources, and a bounded nonempty reason. Gemini uses a structured response schema, but the local validator remains the actual consumer-side check. Source references are citations: they cannot suppress compulsory companion observations, trigger a new acquisition, select arbitrary endpoints, change cadence, edit policy or provide a permission token. The runtime creates `ActorContributionV01` with actual `contribution_mode` `CLOUD_LLM` and `actor_role` `provider_llm`, checks it against the semantic request and default trust profiles, and builds a review packet. Intended local role and actual cloud origin remain separate fields.

The provider response schema is closed:

```
{
  "diagnostic": "ONE_ALLOWED_DIAGNOSTIC",
  "source_refs": ["KNOWN_SUPPLIED_SOURCE_ID"],
  "reason": "A bounded reason for the proposed investigation"
}
```

This example is a schema illustration, not a captured provider response. The actual allowed diagnostic enum depends on the duty. Local and adversarial duties can select `REFERENCE_INTEGRITY`, `INDEPENDENT_RESERVE_SERIES` or `EVIDENCE_GAPS`; the environmental duty can select `TEMPORAL_CONTEXT_APPLICABILITY`, `SPATIAL_CONTEXT_APPLICABILITY` or `EVIDENCE_GAPS`. The runtime checks the returned shape and values even though the provider is also asked for structured JSON.

The role context's `source_refs` meaning is explicitly `CITATIONS_NOT_ACQUISITION_OR_OPTIONAL_SAFETY_FILTERS`. Compulsory companion observations remain in the material whether or not the model cites them. A returned citation therefore cannot remove an unfavorable local witness or authorize a network read. The runtime, rather than the model, chooses the current acquisition boundary when the accepted investigation actually requires one.

6. ActorContribution and BSEP are evidence, not permission

`kernel_adapter_v01.semantic_source` validates the material, builds a business context and orchestrator route context, produces a bounded structured rationale, and constructs the common `BoundedSemanticEvidencePacket`. The BSEP observed facts include a compact selection/input_ref pair and the semantic request, bounded context and contribution identities with their roles. It carries missing-evidence, uncertainty, risk, required-guard and authority-boundary records. BSEP is explicitly neither truth nor an `ActionCommitPacket` nor `FinalOutput`. The full validated semantic material is also bound in the advisory `SemanticArchitectProposal` and actual Work input; a small BSEP does not mean the consumer discards the model contribution. Raw cross-role text is absent from BSEP, while private captures preserve source identity outside the public presentation.

`semantic_adapter_v01.bind` creates a `SemanticWorkRequestV01`, dependency evidence, a normalized claim and an `ActorContributionV01`. Actual Gemini records keep `contribution_mode=CLOUD_LLM` and `actor_role=provider_llm`; the requested local duty is recorded separately. Common semantic validators check the request and contribution against the trust profiles and form the review packet.

The runtime BSEP uses `schema_version=bounded_semantic_evidence_packet_v0.1` and `packet_type=BoundedSemanticEvidencePacket`. Its identities bind the source route, proposal, route context, structured rationale and business request. Its actual recorded authority flags keep `truth_claimed`, `authority_claimed`, `action_permission_claimed`, `action_commit_packet_claimed`, `final_output_claimed`, `root_bypass_claimed` and `real_world_effects_allowed` false. `root_final_authority_preserved` is true.

These flags are not the whole enforcement mechanism. Their claims are consumed together with actual builders, closed material checks, current-source bindings, Root inputs and effect-path validation. A JSON flag saying that authority is absent cannot substitute for those producer-to-consumer checks. The companion `semantic_boundary_trace_v01.json` identifies the actual recorded artifacts and phase relationships, while clearly remaining a new derivative.

7. Common C/D and actual Work materialization

The public `source_family` path creates a local routing snapshot and actual source context, calls `route_execution_mode_v01`, validates the proposal and independently reviews it at Root. It projects route eligibility and builds `build_fractal_runtime_source_context_v02`. The selected first live composition uses `full_fractal` with bounded work. `local_slm` remains unavailable because there is no installed local model. `assemble` creates a real Work candidate, a D review obligation, invokes `run_fractal_runtime_v02`, validates the returned binding, advances the Work program and validates actual results. The pure Work item does not make hidden provider calls: semantic production occurs earlier under its own transport accounting. This is actual common C/D execution, not a domain-specific simulation named after those stages.

The C route is established by `route_execution_mode_v01` and `review_execution_mode_proposal_v01` from a current local snapshot. The D source is created by `build_fractal_runtime_source_context_v02`. `assemble` materializes a finite diagnostic Work item with a real review obligation, invokes `run_fractal_runtime_v02`, validates its result binding and advances Work only after that obligation is satisfied.

The domain's pure capability `sentinel.diagnostic.v01` computes the declared diagnostic and has output validation against the actual computation. Its result passes through ordinary Work result artifacts and independent current Root review. This is why a presentation can say that a model's selection changes work without saying that the model directly executed arbitrary code.

The first live three-role composition is the lane with the actual public D invocation. Later A/B and internal-role continuations use the corresponding bounded current consumers; the evidence does not claim a new full D for every response. Model calls happen at the explicit semantic-production boundary, not inside the pure Work operation's zero-model-call budget.

8. PLAN, reviewed basis, native read and RESULT

The corrected operation has two Work phases. `plan_roles` builds pure PLAN material with no series read. A `ReviewedWorkBasis` binds live Host, program, results, artifact, current dependencies and Root review. `execute_role_plan` checks the entire batch and all current bindings before native acquisition. `READ_RESERVE_SERIES` then requires ordinary current action preparation, Root decision, Host installation and dispatch. Its resulting `EvidenceReceipt` is attached to the later RESULT material; the later Work result is again reviewed by Root. In the latest live trace the series branch has PLAN at phase 10, native read receipt at 15 and RESULT at 16; the offline-site emulation branch repeats the order at phases 18, 23 and 24. Those ordinal relationships are source evidence, not a drawn narrative substituting for execution.

The actual `ReviewedWorkBasis` contains live Host, Work program, results, result artifact, common source context, review bindings, material bytes, dependency bytes and Root review. An exported dictionary of references cannot recreate this live basis. Validation compares the current Root, task, policy, frames, tick, observation set and capability state before the planned read can occur.

For the final live reserve-history arm, recorded phase indices provide a concise witness:

Phase index (zero-based)	Phase	Meaning
10	CURRENT_WORK_ROOT_ACCEPTED / PLAN	Pure plan reviewed under the current basis
11-14	Root preparation, Host install and dispatch	Ordinary bounded action path
15	ACTUAL_REVIEWED_SERIES_READ	Native receipt bound to the accepted plan
16	CURRENT_WORK_ROOT_ACCEPTED / RESULT	Acquired result consumed and reviewed

The internal-role branch after site withdrawal repeats that order at 18, 23 and 24. The saved-response path later consumes the existing receipt at phase 26 with `creates_action=false`; this is separately labelled captured reexecution, not a new native read.

9. The three real semantic contributions

The latest first three actual Gemini responses share one current composition. `LOCAL_SLOPE_ANALYST` selects `REFERENCE_INTEGRITY` and compares current metadata; `CLOUD_ENVIRONMENTAL_ANALYST` selects `SPATIAL_CONTEXT_APPLICABILITY` and checks declared site identity; `ADVERSARIAL_SENSOR_REVIEWER` also selects `REFERENCE_INTEGRITY`, independently contributing its review. There is no requirement that every role select a different enum. The evidence demonstrates contribution identities in the common BSEP, D source context, current PLAN and RESULT materials, outputs and independent Root acceptance. The environmental result supports synthetic site identity while retaining physical_applicability `UNRESOLVED_NO_VERIFIED_MAPPING`. Informational Root acceptance in this composition reports `permission_created false` and `effect_requested false`; it does not sound a siren.

The three first contributions share Root `SiteSafetyRoot:sentinel:xtejghnufwcs`. Their current PLAN Root decision is `643be8eaf78ad74905f95f34ce466b226c8c1fcc90b052cf80b1ff98c7f1159e`; RESULT decision is `f5679c23f0581795dcaaf509596227676cabf60ab238c178264343d7f33f5154`. The PLAN BSEP is `bsep:sentinel:orchestrator:9ca7630b3dca877abb091858766c003d2c5a3fd16e7ae17d431978a7476a0e5a`.

These are historical recorded references, not reusable current authority. The role-local result compares current channel metadata. The environmental result compares synthetic site identity and retains `physical_mapping=NOT_ESTABLISHED`. The reviewer supplies a separate contribution even when choosing the

same enum as the local analyst. The result is a real multi-role composition, not an assertion that agreement between two roles proves independent physical truth.

10. Note-only A/B and the actual measured contrast

The final pair holds non-note context constant and changes only the maintenance note. Note A questions the health and current calibration association attached to the latest reserve record. Note B questions the range and measured duration of the reserve history. Both notes are selected after the implementation freeze and before the live calls. Their non-text envelope remains the same: source identity, site, observed_at 3710, received_at 3715 and version 2. The current observations, policy and operation catalogue are otherwise identical. The experiment therefore tests semantic choice of useful work, rather than a hardcoded numeric threshold or hidden change in source availability. The full note text and byte comparison are retained privately; the presentation paraphrases the bounded uncertainty.

A selects REFERENCE_INTEGRITY and performs record relationship checks over the current displacement/reserve records. It returns supported current calibration associations and distinct lineage identities, without a reserve history read. B selects INDEPENDENT_RESERVE_SERIES. After accepted PLAN and native acquisition, it measures three actual synthetic history samples: span 20 seconds, range 150 micrometres and ELIGIBLE source fitness. The performed_investigation comparator compares checked relationships, source/sample identities and computed measurements; changing a response reason, operation label or hash alone does not establish a contrast. In particular reference and evidence-gap operations performing the same checks count as the same substantive work. This result closes the declared live C01 contrast for this bounded pair.

The current sentinel.investigation.v02 contract names a target, intended use, checked relationship, required evidence and permissible result statuses: SUPPORTED, MISSING, INCOMPATIBLE and UNRESOLVED. This change addresses a real ambiguity. Earlier reserve-history work also checked source independence, so two differently worded questions reasonably selected the same operation. The updated contract distinguishes the purpose of an investigation from the guards needed before trusting its input. Similarly, absence of physical certification is no longer an unconditional evidence-gap answer for every role. It remains an explicit limit, while the actual findings depend on the declared question. The model still has multiple useful choices and can return evidence gaps; no valid response is retried merely to obtain a preferred enum.

The independent pair changes neither the numerical observations nor the available catalogue. The non-note context digest is 27305ed252d2c7b5badd107739ca2c292e6c135c6271c4331f75b47daf1859d0. The note envelope preserves site, source, observation and receipt times and version. The actual series read uses the declared synthetic source synthetic:reserve_series:v01 and its returned sample identities.

Source fitness checks cover admitted source/receipt identity, channel, units, calibration, lineage separation, chronological order, maximum sample gap, latest age and minimum window support. The bounded series policy requires at least three samples, span at least 20 seconds, maximum gap 15 seconds, newest maximum age 10 seconds and range at most 1000 micrometres. The observed final result, 150 micrometres over 20 seconds, is eligible under that synthetic policy. It is not a prediction of future displacement.

The substantive comparator intentionally prevents false wins: different names for the same set of relationship findings do not count as different work. It compares actual performed relationships and measurements rather than prose, random identities or a manually assigned success label.

11. Preserved failures and experiment integrity

Two prior A/B pairs remain NOT_DEMONSTRATED in the public claim map. Their valid answers are not reclassified as errors, discarded, or pooled selectively into a new success rate. LS2R additionally exposed a duty-scoping problem: the evidence-gap description and consumer made missing physical certification a permanent answer, drawing the environmental role away from applicability. LS2R2 changes the domain contracts and consumer checks, freezes the new implementation, then runs a new independent pair. Five actual Gemini attempts in that lane are validated; there are no valid-answer retries. The acceptance claim is the recorded finite contrast and causal consumption, not universal model behavior across arbitrary prompts or probabilistic reliability estimated from a single pair.

The record distinguishes a technical failure from a valid scientific non-result. A provider transport/schema failure may be addressed without treating a valid alternate answer as an error. Earlier valid A/B choices remain inconclusive. The task and executor semantics were repaired before a new frozen experiment; the older outputs were not filtered out.

The environmental-role issue was also traced through both producer and consumer: a permanent physical-certification gap was emphasized in the old operation description and unconditionally returned in code. Removing that universal prerequisite did not invent certification. It made the actual gap findings relative to the investigation's intended use while preserving physical certification as an explicit unestablished claim.

This distinction is essential for reproducibility. The final outcome demonstrates a bounded pair on one recorded model version with retained inputs and responses. It is not a confidence interval for all future models, seeds or maintenance notes. A subsequent change of model or domain contract should retain that provenance rather than inheriting the exact outcome as a universal guarantee.

12. Actual rainfall E and the configuration effect

The chronological story prepares an unexecuted next monitoring configuration. A changed rainfall dependency invalidates that pending packet. The common E path captures the current source and invalidation, selectively recomputes affected D work, retains unchanged sibling evidence, and supplies the actual consumed result to current configuration Work. A fresh current Root decision and native APPLY_MONITORING_CONFIG dispatch then determine the next intake. The previous packet cannot execute again or return from CONSUMED. This proves a meaningful action basis for E before any alarm: configuring how the next local observation will occur. It does not merely generate an explanatory report or claim a new cadence in text.

A separate recorded current-policy experiment sets `rain_fast_mm_x1000` to 50000, rather than the original 30000. Rainfall changes from 12000 to 41000 in fixed-point millimetres. Under the accepted threshold of 50000, the actual E result and configuration Work retain the 60-second cadence. The mock configuration is applied at scenario tick 90 with `due_at` and `next_intake` 150, and the recorded next intake occurs at 150. Under the old threshold, 41000 would select the fast 15-second interval. This is a discriminating current-policy check. The source contract, changed source artifact, recomputed result, retained consumption and final review are all retained. A policy change during an old continuation is refused with `policy_change_requires_fresh_baseline`; changing metadata cannot launder an obsolete D baseline.

The retained current-policy result binds its source contract, source artifact, changed D output, preserved D sibling, consumed binding and current Root review. Its recomputed result reference is `frcellresult_v02:fceb7faeb6165b9427ef9a31b65cc126fd847f6ed1e07ea38944835c3e9359c7`; retained sibling reference is `frcellresult_v02:1c41740e244e6ea4f2b23d231124b81f8f628ac64c93b52c9c7aedc0477a7bff`. `APPLY_MONITORING_CONFIG` returns an actual mock result with `cadence_seconds=60`, `configured_at=90`, `due_at=150` and `next_intake=150`.

The source update and fresh observation do not extend an expired action's permissions. The old pending packet is kept as historical evidence and its refusal is tested. The new cadence comes from new reviewed work under the accepted policy, not from editing the old action object. The unchanged configuration path is connected to

the final source through the accepted source-impact bridge; no new E was run solely for the presentation or main landing.

13. Local action while optional reasoning remains pending

C06 uses a controlled pending provider, not an artificially delayed display of an already returned answer and not a paid timeout claim. The optional worker receives an immutable snapshot and is demonstrably started before the fresh critical observation. It holds no Host lock or effect authority. A fresh local invocation under the same Root and Host consumes current observations and a separately accounted finite budget. Work, Root and native mock ON complete before the worker is released. Offline-site and online-but-nonresponding branches are both retained. The late response keeps its old context identity and does not change the current result automatically. Completing or cancelling the optional request is not a precondition for local action.

The pending-worker demonstration exercises real ordering rather than a staged user-interface illusion. The worker has genuinely entered the optional request and remains waiting. It cannot hold the Host lock while local evidence is processed. An actual fresh local task then obtains current Work and Root checks before native ON. The old request's immutable context is retained and its later response is historical to that context.

The local invocation receives a finite explicitly allocated budget; the earlier optional request's spending remains in the same episode account. C05 separately tests reduced Work budgets and infeasibility. This establishes a bounded coordination pattern: local responsibility can progress even if optional reasoning is delayed. It does not claim preemption of arbitrary provider code, a hardware real-time scheduler or cancellation of every external request.

14. Measured recovery, current policy and distinct OFF

The finite synthetic incident policy requires both displacement and reserve below or equal to 1000 micrometres, at least three new independent pairs, measured coverage of at least 30 seconds on each channel and no gap above 15 seconds. Current policy controls admissible local age. In LS2R2, a policy allowing age two seconds actually rejects five-second-old data; fresh witnesses can then establish valid recovery. Tests cover boundary values two, ten and twelve. Witnesses from changed calibration, lineage, units, statistic or relevant policy cannot be combined with earlier coverage: old witness history is retained as revoked evidence. Receipt timing and repeated old IDs cannot create a recovery interval. Even clearance remains a candidate condition until an independent current Root decision and native OFF complete.

`Incident.evaluate` calls the module-level `features` function, which passes the current age limit through `local_pair` into `EventBook.current`, rather than accepting a policy value and ignoring it in the recovery path. The corrected tests discriminate an accepted age of two seconds from the old fixed ten-second behavior. Fresh data at the permitted boundary can support recovery; stale data beyond that policy cannot.

Recovery coverage is keyed by relevant source and policy facts: site, channel, lineage, calibration, unit, statistic, recovery policy and the current critical/age parameters. Observation IDs themselves do not become the stable dependency fingerprint; new compatible samples must be allowed to accumulate. A change of stable dependencies revokes old coverage while retaining it as historical evidence. Repeated old IDs, nonadvancing measured intervals and excessive gaps do not accumulate the required duration.

The resulting `clearance` condition is not the OFF effect. The local coordinator still performs a normal current decision and native dispatch, and only the resulting receipt closes the incident. Cleanup, late LLM text, a recovered network or a delivered report has no equivalent authority.

15. Immutable outbox and acknowledged delivery

The local mock outbox creates an immutable report payload, message identity and payload digest. Delivery attempts preserve those exact bytes, and the receiver acknowledgement must match the same identity and hash. The retained story records QUEUED, ATTEMPTED and RECEIVED. Negative controls cover a missing acknowledgement, changed payload, receipt misuse and expired action packets. A receipt is evidence of a prior bounded action; it is not authority to send again. This demonstrates a local source-bound report protocol and exact acknowledgement semantics. It does not claim distributed exactly-once delivery over arbitrary networks or that a public emergency authority received an alert. Network recovery does not clear the incident; recovery/OFF is a separate local path.

The report's payload identity is derived from its exact bytes. A receiver must acknowledge the same message and payload hash. Queue state and delivery state are therefore separate from the incident's safety state. A lost acknowledgement leaves an explicitly uncertain outcome instead of a made-up success. This local protocol can be inspected in `outbox_v01.py` and the historical C09 records.

The package preserves failed or refused operations as evidence. It does not equate a previously recorded receipt with a fresh permission to invoke an effect. Distributed exactly-once delivery, durable transport across arbitrary failures and external agency acknowledgement are beyond this local mock demonstration.

16. Dual profiles and truthful engine provenance

The low-cost architectural profile performs deterministic local algorithms and local semantic-memory descent with no model transport. The richer architectural profile includes an internal analyst intended for a local SLM, such as a model-capable compact edge computer. No such physical local SLM is installed or benchmarked in this demonstration. Actual Gemini cloud transport emulates that role through an independently authorized harness. After site-cloud and remote-rainfall withdrawal, the fifth real model call still serves only the internal local duty; it remains CLOUD_LLM/provider_llm with physical_local_slm unavailable. When harness transport also becomes unavailable, the role cannot be relabelled as functioning local hardware. Hardware price, memory fit, model parameter count and power usage are deployment possibilities, not measured results here.

The fifth latest live contribution is the useful boundary case. Its capability snapshot says site-cloud unavailable and remote rainfall unavailable, while the separately authorized harness transport remains available. Its intended duty is LOCAL_SLOPE_ANALYST, its actual mode remains CLOUD_LLM and physical local SLM remains unavailable. Thus modeled site outage does not deceptively erase real provider traffic.

The genuinely model-free branch disables the model route and performs fresh local work and LocalDRS descent with zero model transports. These are two architectural profiles under common authority boundaries. The richer node is a plausible future compact-computer deployment with a local model, but no price, model-size fit, power, throughput or Mac hardware result was measured. Role substitution is demonstrated semantically; physical hardware substitution is an explicit future integration.

17. LocalDRS and captured reexecution

The runtime uses actual LocalDRS semantic records, queries, resolution and descent. A semantic recipe or incident summary can be written after appropriate local review. Warm selection evaluates current query scope, root identity, dependency fingerprint and time, and requires fresh Root review before informational descent. Policy or calibration changes make stale content ineligible; old data remains history instead of becoming the current plan. The latest fresh LOCAL_ALGORITHMS_ONLY continuation records one search, one descent, two local Work operations and zero model transports, with a fresh local intake. This is different from CAPTURED_REEXECUTION, which explicitly consumes a saved semantic response through current consumers and retains original provenance. Neither operation is an executable permission cache.

Three mechanisms must stay separate:

Mechanism	Inputs	New authority treatment
LocalDRS	Reviewed semantic record selected by current query/dependencies	Fresh eligibility and Root review before informational descent
Captured reexecution	Exact saved semantic response and recorded receipt	Current consumers; no claim of a new provider or read
Safe package replay	Frozen derivative bytes and independent pin	Evidence-only verification; no live Host restored

The latest fresh local-only branch records one sensor intake boundary, two local Work operations, one DRS search and one descent, with zero model transports or effects. The captured branch records two local Work operations and zero provider, sensor, D/E and effect calls. These named counters are scoped instrumentation, not an OS-wide network isolation proof.

18. Real EA acquisition and exact unknowns

Historical Environment Agency acquisition first handles station metadata before retrieving measurements. Candidate station selection is based on the declared metadata/recency rules, with the chosen source pinned before its values are read. The successful retained source is 4163, Day Brook. Eight actual rainfall records were normalized; each reported value is zero. This does not fabricate a storm, replace missing records with zeros or pick a different station after inspecting values. The adapter retains unknown interval endpoints. It therefore does not aggregate these readings into a supported current hourly total or claim physical mapping from the UK gauge to slope_demo_site_001. A prior metadata-shape failure and the prior LS2 empty-reading result remain separate historical records. This slide is historical data evidence, not a current weather report.

The current temporal investigation explicitly tests scenario-second ordering start <= end <= received <= evaluation; hourly accumulation requires a 3600-second interval, unit mm_x1000, statistic accumulation_1h and an adequate record. An old endpoint remains historical; a point labelled as an hour is incompatible. Missing interval support is named rather than invented. Spatial applicability separately compares declared synthetic site identities and always preserves the absence of a verified physical mapping. The real live environmental selection in LS2R2 performed the spatial investigation, not a live temporal choice: temporal fixes are established by targeted current consumer checks. A bounded SUPPORTED_FOR_DECLARED_USE result can coexist with UNRESOLVED_NO_VERIFIED_MAPPING, because those are different claims.

The real acquisition and the synthetic rainfall applicability consumer do not use interchangeable clocks. EA records carry real source timestamps and unknown interval endpoints; the current synthetic hourly consumer expects explicit scenario seconds and a supported 3600-second interval. The adapter does not silently assign synthetic times to the real data, infer an hour from a point timestamp or map a UK station onto the synthetic slope.

This refusal to overinterpret is part of the useful result. The system can acquire real records, normalize valid zeros and still state that the evidence cannot support a requested physical or interval claim. The latest live environmental operation chose spatial applicability, so its actual successful work is synthetic identity comparison with unresolved physical mapping. Current temporal correctness is separately supported by targeted producer-to-consumer checks.

19. Ten named contrast requirements

The public claim map indexes exact source archives, members, sizes and hashes. C01 is the latest live note-dependent substantive-work contrast, with earlier inconclusive pairs retained. C02 sends old observations with new receipt times through actual applicability Work and Root. C03 removes primary and reserve evidence and observes bounded insufficiency rather than fabricated action. C04 challenges typed time, identity, projection, sequence and independent lineage. C05 reduces the available Work budget from two to one to infeasible zero within the same episode budget. These are not five newly rerun post-landing cases: each row carries its execution lane and later source-impact disposition. A test exclusion in default pytest is neither a pass nor an execution.

C06 demonstrates a fresh local invocation while an older provider worker remains pending. C07 requires measured recovery and a distinct OFF, with subsequent fixes for source/policy changes and actual accepted local age. C08 admits additional synthetic channel-level and vibration evidence under the same site Root. C09 checks immutable report/acknowledgement identity and refuses receipt-as-permission or expired-packet reuse. C10 exercises current policy/calibration, stale DRS eligibility and actual rainfall E with consumed results and retained siblings. The latest package does not cryptographically anchor all these historical rows; the source-bound claim map describes their provenance and limitations. The whole-source admission later preserves the accepted implementation with narrow setup-alignment test corrections.

The published `claim_evidence_map_v01.json` supplies exact archive, member, byte length and SHA256 for each cited record. C01-C10 and S0-S5 are not represented as a fictional single fresh test suite. Their `execution_mode`, original outcome, later correction and current applicability remain visible. The two old inconclusive live pairs are retained beside the later live result.

Default pytest excludes three explicitly invoked Sentinel audit modules, preserving the pre-existing EWS exclusions. An excluded test is neither a passed test nor a claim that its historical setup still matches every later contract. Landing aligns two narrow setup examples and records their actual fresh checks. This preserves the distinction between current source correctness, old execution evidence and the owner-facing published account.

20. Public package, independent pin and replay

The public three-file package consists of `projection.json`, `manifest.json` and `publication.json`. It covers only the enumerated LS2R2 safe-derived fields. It does not anchor the entire S0-S5 history, all C01-C10 cases, earlier E or EA acquisition. Those claims are linked through a separately identified source-bound index. The original package keeps its EVIDENCE_ONLY, TEST_SUPPLIED_PIN and PENDING_REVIEW labels; later independent lead review and detached verification are separate records. Public presentation assets do not silently become members of the old seal. The landed showcase excludes raw provider captures. The new reader separately includes reviewed bounded synthetic inputs and exact structured outputs, with explicit provenance and no credential or private-environment dump. A self-contained LLM reader can include safe explanation, exact public evidence, code and clearly labelled derived historical summaries without impersonating private original evidence.

The verifier reopens the package and receives the expected anchor externally. The accepted pin is `7b91c496521e11c69fab9ea89a576474d37dbcb93c494b58a93269884fffb9b8`. Genuine safe-derived replay passes; wrong-pin, changed-byte and coherent causal-link substitution controls are refused. Instrumented named boundaries record no new model, sensor or consequential effect calls during replay. The public wrapper uses an existing interpreter and dependencies and performs no installation. ANCHORED_PASS means the exact supported derivative is consistent with the independently supplied identity. It does not certify a sensor, authenticate a human signer, establish a trusted timestamp, reconstruct a live Host, create action permission or canonically replay unsupported historical native graphs.

The expected independent pin is:

```
7b91c496521e11c69fab9ea89a576474d37dbcb93c494b58a93269884fffb9b8
```

The existing public verification command is:

```
sh docs/showcase/landslide_sentinel_v01/verify_replay.sh 7b91c496521e11c69fab9ea89a576474d37dbcb93c494b58a93269884fffb9b8
```

SENTINEL_PYTHON may select an existing interpreter; the wrapper otherwise uses the checkout's .venv/bin/python or python3. It does not install dependencies or run the demonstration. This command is supplied for inspection, not executed as part of creating the presentation.

The immutable package keeps its original evidence-only and pending-review fields. The later detached lead review and verification record are a different event, not a rewrite of those bytes or an authenticated signature. Full canonical replay of unsupported historical native graphs remains UNSUPPORTED_NATIVE_SCHEMA. A native DTO projection may preserve source identity while still being a derivative, rather than the original graph.

21. Timings, budgets and repository admission

Latest LS2R2 live stage: 205.701933 seconds, including one common D of 100.552787 seconds and five Gemini calls totalling 67.423857 seconds. A distinct earlier LS2R current-policy experiment takes 381.194265 seconds, including common D 99.504709, common E 242.586459, supplied D 11.892811 and supplied E 13.105713. These timings overlap within their respective stages and must not be summed into one mega-run. The stage is not the coding assistant's wall-clock completion time. No per-reading LLM call, full D/E per sensor event or certified deadline is asserted. The finite local pending-provider path is separately demonstrated by ordering, not by borrowing the D/E duration as an alarm-response measurement.

The source landing commit is 2f328be634247be11bc18a3b22a919f393d6ed1d. The owner return records ordinary push, matching remote branch and clean working tree. The two setup-alignment tests were corrected for the accepted investigation/time rules; their actual new checks remain separate from prior runtime evidence. The admission matrix preserves nine existing Living registrations and uses the established release binder rather than inventing a tenth authority or new Gate. No fresh Gemini, D/E or EA run was required merely to publish accepted source. The presentation is a subsequent documentation/artifact operation and must not rewrite the frozen accepted evidence. Source acceptance for this synthetic demonstration does not amount to scientific certification or physical deployment readiness.

Landing details: final admitted state SENTINEL_IMPLEMENTATION_ADMITTED_COMMITTED. The 46 landed postimages plus 1,015 protected base files produce 1,061 final tracked entries. Two aligned tests passed in 11.82 seconds; the final three guard/release checks took 72.92 seconds; one Living-registry check took 7.37 seconds and retained exactly nine registrations. Across retained landing executions there are nine successful CALLs over six distinct test functions, not a single nine-test suite. Owner replay took 0.399 seconds and reported ANCHORED_PASS with zero new named execution calls. These are read from the detached landing return, not new executions for this presentation.

Recorded scope	Time	Scope note
LS2R2 live stage	205.701933 s	Includes five provider calls and one D
That stage's D	100.552787 s	Included in the stage above
Five provider durations	67.423857 s	Included in the same live stage
LS2R current-policy stage	381.194265 s	A different recorded experiment
Its D / E	99.504709 / 242.586459 s	Included in its stage
Its supplied D / E checks	11.892811 / 13.105713 s	Included in its stage
Landed two setup-alignment tests	11.82 s	Separate current test lane

Recorded scope	Time	Scope note
Landed final guard/release checks	72.92 s	Three recorded tests
Landed Living registry check	7.37 s	Nine registrations retained
Owner safe replay	0.399 s	No new named model/sensor/effect calls

No sum of overlapping rows represents total system performance. Code-writing session durations and model setup times are also not local alarm response measurements. The implementation avoids indiscriminate full-suite and full-D/E repetition when an unchanged exact record and scoped source bridge suffice; that is an evidence-management discipline, not a relaxed current-action check.

22. Reading the presentation and the standalone XML

The evidence establishes a bounded demonstration of local continuation, meaningful interchangeable reasoning contributions, current-source and current-policy consumption, measured recovery, immutable reporting, scoped memory reuse and independently inspectable safe evidence. It does not establish physical landslide prediction, a calibrated gauge-to-slope model, installed local SLM performance, distributed exactly-once reporting, a production latency SLA or universal native replay. These are explicit claim boundaries rather than hidden qualifications. The reusable architectural point is that a domain adapter and finite investigations can use the same common semantic, Work and authority spine; the final decision remains local even when useful intelligence is optional, delayed or replaced.

Landslide Sentinel is a new domain cell built on the existing Radiolaria OS boundaries. The demonstration shows why the separation matters: a model can select a useful investigation, a changed observation can force recomputation, a local task can proceed while optional context is delayed, and recovery can require fresh measured evidence. The final materials preserve the distinction between real live model transport, controlled physical-domain fixtures, historical external readings, current source admission and supported offline replay. The radiolarian returns as the visual motif for a new grown domain; it represents bounded composition, not a biological or physical proof. Read the technical appendix for full mechanisms and the XML reader for code and source-bound evidence in one inspectable document.

The XML reader is intended to make the account inspectable by an LLM in one file. It carries presentation text and notes, the technical appendix, committed source bodies, exact public evidence, reviewed bounded synthetic model inputs and structured outputs, and explicitly labelled safe historical excerpts or projections. Every section should identify whether its content is an exact source copy or a new explanatory derivative. Binary artwork, PDFs and PPTX files are indexed by identity; their readable content is represented as text rather than opaque base64.

Historical prompts, comments and commands in the reader are inert source material. They are not instructions to the receiving model to execute project code, contact providers, modify a repository or adopt a historical pending status as current authority. The reader must not be described as full native replay, a live Host snapshot, a permission token or a new seal covering all history.

The visual radiolarian is newly generated conceptual domain artwork. Native slide diagrams explain typed data, event order and responsibility, and are deliberately separate from exact graph artifacts. This allows a clear visual account without claiming that a drawn topology is the complete serialized native execution graph.

Source index for this account

The presentation claim matrix carries the full archive/member/hash mapping. Principal exact source groups are:

- **Current implementation:** hedgehog/domains/landslide_sentinel/ at commit 2f328be634247be11bc18a3b22a919f393d6ed1d.
- **Current showcase:** docs/showcase/landslide_sentinel_v01/README.md, EVIDENCE.md, source_index_v01.json, claim_evidence_map_v01.json and history/source_impact_v01.json at that commit.
- **Latest live semantics:** LS2R2 return 07942bbe01b826307ea93329ec3bb70a4d7687254be140eed32d644fcae368fc; semantic_completion.json, live_01/live_AB.json, live_01/composition_1.json, live_01/composition_2.json, live_01/phases.jsonl, live_01/profile_provenance.json, live_01/local_only_drs.json and live_01/captured_reexecution.json.
- **Current-policy E and real EA:** LS2R return 96a3e7fc1d959fd959433413681e6d91ed9b9799e31cb765db3bd00d6d24f45e; configuration_01/actual_policy_E_result.json and ea_capture_02/result.json.
- **Controlled history and contrasts:** LS2 return c26ec2bad672071ba05ef597b5c64e5f00f8d5cc83da37f4d8c7b057ed526ef9; exact run members are enumerated by the committed claim map. The cited whole story is runs/20260915T001421055364Z/story/story.json.
- **Older baseline:** LS1 and LS0 identities, original failures, repairs and retained results are preserved in the same public source-bound claim map.

The stated hashes locate evidence; they do not prove physical truth by themselves. This appendix was prepared from saved sources and records without new model, D/E, sensor or physical-effect execution.

23. Research context and executable criteria

Research-informed scenarios. Source-bound execution evidence. Published operational challenges informed the scenario design. Radiolaria demonstrates selected software responses on controlled observations and mock effects; the cited researchers and institutions are not presented as project endorsers.

The chain is published problem > our software criterion > observed Sentinel behavior > exact execution source. R01-R12 establish external context; S and C identify recorded scenarios and contrasts; D01-D07 locate internal evidence. The translation into requirements is our engineering judgment, not a request attributed to the cited authors. Human direction, prepared AI assistants, Codex, domain adapters and the existing kernel all contributed. This is not automatic compilation of articles by BSEP. Specialists can inspect and challenge the mapping; a specialist-led revision cycle remains a future possibility, not an experiment already conducted.

Published challenge > criterion > recorded behavior

The rows below are engineering mappings. Research IDs link to the bibliography; internal D IDs point to records, not endorsements.

# / context	Published challenge	Our software criterion	Recorded S / C / D evidence	Limit
1 / R01, R02	Local hydrology and multiple observations matter to warning decisions.	Keep source purpose and dependencies; do not collapse the task to one rainfall value.	S1 / CURRENT_POLICY_E consumes changed monitoring configuration and cadence; C08 admits another evidence profile. [D02, D04]	Software dependency handling, not the authors' hydrologic model or geological accuracy.
2 / R04	Storms, power loss and poor visibility can interrupt monitoring.	Distinguish capability failure, usable reserve and insufficient observations.	C03 records LIMITED after primary loss and INSUFFICIENT after reserve loss; S3 permits bounded local continuation. [D02, D05]	Controlled inputs; no reproduction of Bertorelle et al.'s virtual-sensor model.
3 / R03	Operator experience highlights robust monitoring and redundancy.	Check provenance and independence, not just the presence of a second message.	C04 rejects duplicate, misordered or shared-upstream evidence as independent confirmation. [D02, D03]	Declared lineage checks cannot independently establish physical truth.
4 / R05	Debris-flow vibrations must be distinguished from other signals.	Use ambiguity to select a bounded additional investigation.	S2/C01's latest live pair changes only the source-bound note; actual Work checks metadata or reads and processes reserve history. [D03]	Meaning affects executed work; no seismic ML detector or claimed superiority.
5 / R06	Post-fire debris flows may leave inadequate warning time after local detection.	Avoid mandatory dependence on optional cloud reasoning once local action is authorized.	C06/S3 performs fresh local Work through the same Root/Host and mock ON while the controlled provider worker remains pending. [D02]	Our timeliness criterion; no forecast lead-time result or field deadline.

# / context	Published challenge	Our software criterion	Recorded S / C / D evidence	Limit
6 / R07	A measurement interval and its telemetry delivery frequency differ.	Separate observed and received time; check interval, statistic, units and purpose.	C02 rejects fresh arrival as proof of freshness; later checks require supported hourly intervals. Real EA capture preserves unknown endpoints. [D02, D03, D06]	HTTP success establishes neither current slope evidence nor a verified gauge-to-site mapping.
7 / R08	Open review asks for clarity about temporal resolution and operator-dependent interpretation.	Retain source-bound notes, versions, windows and inputs actually used.	C01/C02/C04/C10 retain the evidence and conditions behind investigation, applicability and policy decisions. [D02, D03]	Process inspectability; no inter-operator study, ROC/AUC evaluation or judgment that final article errors remain.
8 / R09	Delayed soil-moisture products can remain useful as prior context.	Distinguish historical usefulness from eligibility for a current action.	C02 preserves context-only data; C10/LocalDRS makes stale policy or calibration dependencies ineligible for current reuse. [D02, D05]	Conditional memory reuse; the authors' ERA5-Land model was not executed.
9 / R10	Multiple surges complicate the duration of an alarm.	Require positive recovery evidence; quiet, missing messages and expiry alone cannot clear an incident.	C07/S5 uses a fresh measured sequence and a separate native OFF; repeated frames and cleanup do not establish recovery. [D02, D03]	Synthetic recovery rules; operational thresholds still require domain expertise.
10 / R11, R12	Observation, warning dissemination and human response are distinct; alert messages have a lifecycle.	Separate incident state, signal, outbox delivery and acknowledgement.	S4/C09 retains an immutable report, fresh delivery, matching identity/hash and UNKNOWN after missing acknowledgement, without repeating ON. [D02]	Local mock receiver; CAP-inspired clarity, not CAP conformance, mass warning or demonstrated evacuation.

Time and authorship

Approximately 12 hours of Sentinel implementation on the existing kernel - owner-reported elapsed time. The owner's estimate covers implementation and checks reaching accepted commit/push. Earlier kernel/domain development is excluded; the complete duration and cost of prior research and specification are not established. Presentation preparation followed implementation and was interrupted by tool limits. This is neither an independently timed benchmark nor a comparison against another stack, a model-call duration, an alarm-response measurement or total research-and-development time.

Whitepaper paragraph

Landslide Sentinel demonstrates a path from published domain challenges to executable software criteria. Reports of monitoring failures, delayed telemetry and multi-surge alarm handling informed selected requirements and counterexamples. Built on the existing Radiolaria kernel, the demonstration connects these responses to local decisions, actual result consumption and inspectable execution records. Its value is a rapidly constructed, traceable prototype: published context, explicit requirement, implementation and recorded evidence remain distinguishable and linked. The owner reports approximately 12 hours for Sentinel implementation on the established foundation; this is not total research-and-development time or a comparative benchmark. The case supports a practical direction for testing software responses to domain requirements, not a claim of completed geophysical validation, field readiness or measured scaling performance.

Bindings: R04/R07/R10 establish published context; D01 establishes the preserved foundation; D02-D06 locate behavior. Timing comes from the owner's account in handoff V02 §3.2.

Evidence boundaries

S0-S5 and C01-C10 span controlled, live and later correction lanes. They are not a newly executed aggregate test suite. Earlier inconclusive C01 pairs remain history beside the later valid live pair. The real cloud provider's provenance remains visible; a physical local SLM was not installed. A model-free local branch is distinct from cloud emulation of a local analyst role. The independent safe-derived pin covers only enumerated LS2R2 package fields, not this research insert, the whole history or physical truth. No domain, provider, sensor or D/E execution was performed for this update.

Research references identify published context, not collaboration or endorsement. The scenario mapping and software implementation are Radiolaria's own. Scientific performance reported by cited authors is not attributed to this demonstration. All external descriptions are short original paraphrases with bibliographic attribution; no third-party figures, photographs, logos or long quotations are reproduced. The source collection spans different years and countries and establishes no causal narrative about a particular Nepal disaster. No scientists' participation in the project is implied.

Bibliography

Primary pages checked on 21 September 2026. Rights statements distinguish the source from permission to reuse every embedded asset.

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R05 - Michael Keller. [AI provides debris flow warnings](#). ETH Zürich; 2021-03-26. Institutional account names Małgorzata Chmiel and Fabian Walter. Linked research: Chmiel, M.; Walter, F.; Wenner, M.; Zhang, Z.; McArdell, B. W.; Hibert, C., Machine Learning Improves Debris Flow Warning, Geophysical Research Letters (2021). Rights: No reusable asset licence established for this deliverable; bibliography and own paraphrase only. [Related research](#).

R06 - USGS Communications and Publishing. [PubTalk 10/2018 - Post-fire debris flow early warning.](#)

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R09 - Nunziarita Palazzolo; Antonino Cancelliere; Robert D. Zofei; David J. Peres. [Use of delayed ERA5-Land soil moisture products for improving landslide early warning.](#) NHESS 25, 4907-4919; 2025-12-10. University of Catania study; its scientific results are not Sentinel results. Rights: CC BY 4.0 [Licence terms.](#) [DOI.](#)

R10 - Alexandre Badoux; Christoph Graf; Jakob Rhyner; Richard Kuntner; Brian W. McArdell. [A debris-flow alarm system for the Alpine Illgraben catchment: design and performance.](#) Natural Hazards 49, 517-539; 2008-11-07 online; June 2009 issue. Publisher's open abstract inspected, not subscription full text. Describes an implemented and operating alarm system. Rights: Subscription article; no open reuse licence asserted. [DOI.](#)

R11 - World Meteorological Organization. [WMO and the Early Warnings for All Initiative.](#) undated; accessed 2026-09-21. Four-pillar framework; no initiative membership or implementation of every pillar claimed. Rights: No blanket reuse licence established here; own paraphrase only.

R12 - OASIS Emergency Management Technical Committee. [Common Alerting Protocol Version 1.2.](#) OASIS Standard; 2010-07-01. Sections 1.3.1, 2.2 and 3.2.1 inform lifecycle distinctions; no normative text or dictionary reproduced. Rights: Copyright OASIS 2010; reproduction and explanatory derivatives subject to the document's notice conditions.

Internal evidence key

- **D01:** MAIN landing return > FINAL_REPORT.md: acceptance, preserved foundation and commit 2f328be634247be11bc18a3b22a919f393d6ed1d.
- **D02:** Committed showcase README.md, EVIDENCE.md, claim_evidence_map_v01.json and history/source_impact_v01.json: exact historical records and later applicability.
- **D03:** LS2R2 return > closure_matrix.json, semantic_completion.json and live_01/live_AB.json: latest semantics, provenance and scoped corrections.
- **D04:** LS2R2 return > configuration_01/actual_policy_E_result.json: changed configuration actually consumed.
- **D05:** LS2R2 return > live_01/local_only_drs.json and live_01/captured_reexecution.json: distinct fresh local and captured-response paths.
- **D06:** LS2R2 return > ea_capture_02/result.json: eight normalized records from source 4163, real zeros, unknown endpoints and unverified physical mapping.
- **D07:** Committed public_safe_package plus lead_reviewed_anchor_v01.json, anchor_verification_v01.json and verify_replay.sh: bounded independently pinned derivative.

Exact archive/member/byte/hash identities and execution lanes remain in the existing presentation claim matrix. This context file is a new explanatory derivative and does not enlarge the frozen replay package.